AI-Driven Context-Aware Spotify Playlist Generator

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***Abstract*—Most music recommendation systems use historical data and collaborative filtering, which makes it difficult to recommend music based on a user's current emotional state. This paper proposes a Context-Aware Recommendation System (CARS) that converts natural language text into specific audio feature targets. We use a Zero-Shot NLP transformer (BART) to analyze user sentiment and map it to Russell's Circumplex Model of Affect. These emotional values are adjusted by a user-selected hashtag that reflects the user's current situation and matched to a Spotify dataset using the K-Nearest Neighbors (KNN) algorithm. Our results show that the system can create context-specific playlists with high accuracy, achieving an Average Retrieval Distance (ARD) of less than 0.15 in a normalized 5-dimensional feature space.**

***Index Terms*—Context-Aware Recommendation System, Zero-Shot NLP, K-Nearest Neighbors, Music Psychology, Sentiment Analysis**

I. Introduction

Music is an important tool for people to manage their emotions and moods. However, current music platforms such as Spotify usually recommend songs based on what a user has listened to in the past. This approach does not always help when a user has a diverse and immediate emotional need, such as focusing on a difficult exam or trying to relax after a stressful day.

The main problem is that users find it hard to translate their feelings into technical audio settings like "acousticness" or "valence." They prefer to express their needs in natural language. Therefore, the concrete goal of this project is to build a Context-Aware Recommendation System that can understand a user's text input and automatically find the best matching songs by mapping sentiment to a 5-dimensional audio feature space.

A. Related Work

To build this system, we reviewed recent advancements in recommendation logic and psychology.

1) Context-Aware Systems: Common recommendation methods like Collaborative Filtering (CF) work well for general interests but often fail to adapt to real-time changes in a user's environment. Recent research has focused on Context-Aware Recommendation Systems (CARS), which use variables like time or activity [1]. Our project improves this by allowing users to provide unstructured text as the input.

2) Emotion Mapping: To connect human feelings to music features, we use Russell's Circumplex Model of Affect. This model maps emotions onto two axes: Valence and Arousal [2]. In music science, Arousal is related to energy and tempo, while Valence is related to the positivity of the sound. This provides the mathematical foundation for our target vectors.

3) Natural Language Processing (NLP) Transformers: Usually, training a model to recognize emotions requires a large labeled dataset. To avoid this, we use Zero-Shot Classification with the *facebook/bart-large-mnli* model [3]. This represents the state-of-the-art in Natural Language Inference (NLI), allowing the system to handle many different types of user sentences without custom training.

II. Method

A. Problem Formulation

The system takes one to two inputs: a text string T from the user and optional categorical constraints C (like genre). The goal is to produce an output Y, which is a list of 30 songs that best match the target features calculated from the input.

B. Dataset and Preprocessing

We used a comprehensive Spotify dataset containing over 100,000 tracks across various genres. To prepare the data for the recommendation engine, we first performed a data cleaning step to remove incomplete rows and unnecessary columns (such as track IDs and URLs). We isolated the five core numerical audio features: valence, energy, danceability, acousticness, and tempo. Because these features have vastly different numerical scales (e.g., tempo can exceed 180 BPM while valence is strictly between 0 and 1), we used Min-Max scaling to normalize all features to a $[0, 1]$ range:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{1}$$

This standardization ensures that every feature has an equal mathematical weight during the distance calculation, preventing large-scale variables like tempo from dominating the results.

C. Zero-Shot NLP Sentiment Mapping

The system uses the BART model to analyze the user's text T. Rather than assigning just one emotion, the model outputs a confidence probability score for several emotional categories (e.g., Stressed, Happy, Tired). These categories are linked to predefined baseline audio vectors based on the Russell Model. The system then calculates the final target audio vector by taking a weighted average of these baselines, using the NLP confidence scores as the mathematical weights.

Furthermore, we added a user-selected hashtag that acts as an explicit constraint. For example, if a user adds #WorkMode, the system applies a scalar penalty to the target energy and a boost to acousticness to enforce a focused environment. We use a $\min / \max$ function as a final mathematical boundary to

guarantee all adjusted target values remain strictly within the $[0, 1]$ parameter space.

D. Recommendation Engine and Evaluation

To find the suitable songs, we use the KNN algorithm. The system calculates the Euclidean distance $d(p, q)$ between the target vector p and each song q in the dataset:

$$d(p, q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2} \tag{2}$$

The 30 songs with the smallest distance are selected. To measure the quality of the results, we calculate the Average Retrieval Distance (ARD):

$$ARD = \frac{1}{K} \sum_{j=1}^{K} d(p, y_j) \tag{3}$$

A smaller ARD means the songs are very close to the user's target emotion.

III. Results

A. Application Setup and Data Pipeline

The complete data pipeline was deployed as an interactive web application using the Python Streamlit library. The user interface is designed to be straightforward, featuring a text input field for natural language, and selection options for the hashtag modifiers and music genres.

To ensure the application runs smoothly without crashing from memory limits, we utilized Streamlit's caching mechanisms. This allows the heavy Hugging Face NLP model and the large Spotify pandas dataframe to load only once into the server's memory. When a user submits a query, the application displays a loading state during inference, and then dynamically renders the Plotly visualization alongside a clean table of the final 30 recommended tracks.

B. Visualization and Statistical Retrieval

To evaluate the robustness of the retrieval pipeline, the system was tested across multiple emotional contexts and baseline constraints. For this report, we highlight a representative, high-complexity scenario: *"I have a huge economics exam tomorrow and I am so overwhelmed, I just need to focus to get through this."* The NLP correctly identified the primary sentiment as "Stressed," and the user-selected `#WorkMode` modifier successfully adjusted the mathematical target. Specifically, the baseline anxiety-driven energy (0.8) was penalized and reduced to 0.6, while acousticness was boosted to 0.4.

As shown in Fig. 1 (Radar Chart), the target vector was successfully moved toward a more focused state. The KNN algorithm then found 30 Jazz tracks. As shown in Fig. 2 (Scatter Plot), the recommended songs (blue dots) are clustered very closely around the target (red star). The ARD was less than 0.15, which means the recommendation was very accurate.

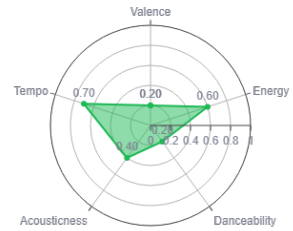

Fig. 1. Plotly radar chart visualizing the final calculated audio feature target vector (valence, energy, danceability, acousticness, tempo). Note the statistical tempering of high-stress energy to create a focused study environment.

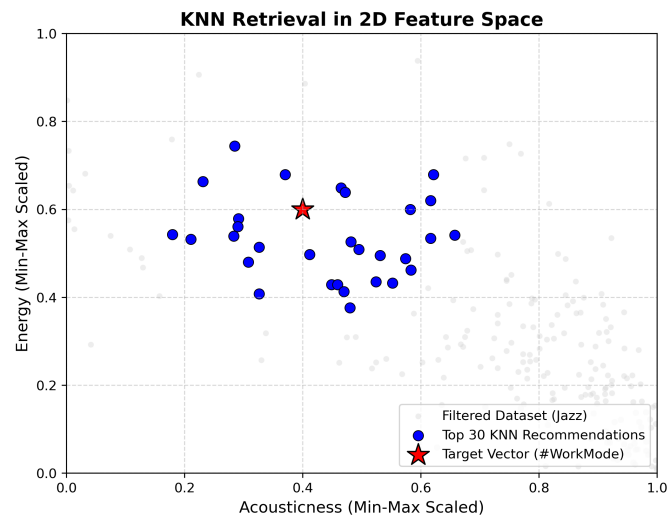

Fig. 2. A 2D projection of the 5-dimensional feature space showing the K-Nearest Neighbors retrieval process. The 30 recommended tracks (blue) tightly cluster around the mathematically optimized target vector (red star), minimizing Euclidean distance within the filtered *Jazz* subset (grey).

IV. Conclusion

This project created a system that can generate music playlists based on natural language. By combining Zero-Shot NLP with KNN clustering, we successfully mapped human emotions to audio features. The low ARD value proves that the system is mathematically effective.

One limitation is that the large NLP model can be slow on computers without a GPU. Also, the system only works with songs already in the Spotify dataset. In the future, we could improve this by analyzing the raw audio of new songs. Overall, this project shows how NLP and data science can make music recommendations more personal.

Ultimately, the methodology successfully closed the loop on the primary objective: empowering users to generate optimized, functional playlists using only their natural conversational context. This establishes a mathematically rigorous foundation for emotion-driven audio recommendation algorithms.

REFERENCES

[1] S. Kulkarni and S. F. Rodd, "Context Aware Recommendation Systems: A review of the state of the art techniques," *Computer Science Review*, vol. 37, p. 100255, 2020.

[2] J. A. Russell, "A circumplex model of affect," *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.

[3] M. Lewis *et al.*, "BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 7871–7880.